

Linear Programming – I: Introduction to Linear Programming Problem (LPP), Formulation of the problem, Graphical method, General, Canonical and standard forms of LPP, Simplex method, Big-M method, Two-phase simplex method.

- **Linear programming** is a method that enables us to identify the optimal solution for a given problem.
- An optimal solution represents the best possible outcome for a specific problem, such as least cost, highest margin, or least time.
- **Example:** Manufacturing issues involve determining the optimal number of units to produce or sell, aiming to maximize profits. This challenge arises when each product requires a fixed amount of manpower, machine hours, and raw materials.
- **Formation of linear programming problem**, involves identifying:
 - **Decision Variables:**

Variables such as x and y that determine the outcome of the linear programming problem, representing the final solution.
 - **Objective Function:**

Typically denoted as Z , it is a linear function that needs optimization based on given conditions to achieve the final solution. Coefficients of the decision variables in the objective function are also called as **costs**.
 - **Constraints:**

Restrictions imposed on decision variables, limiting their values and influencing the feasible solutions.

- **Non-Negative Restrictions:**

Imposed conditions that ensure decision variables (like x and y) are non-negative, reflecting real-world limitations.

Note: A **feasible solution** of an LPP satisfies all the constraints, while an **optimal solution** is a special case of a feasible solution that either maximizes or minimizes the objective function.

The general form of a linear programming problem is expressed as:

Objective Function: $Z = ax + by$

Constraints: $cx + dy \geq e, px + qy \leq r$

Non-Negative Restrictions: $x \geq 0, y \geq 0$

Here, x and y serve as the decision variables in the given conditions.

Linear Programming Graphical Method

- In the process of solving linear programming problems (LPP), we begin by graphically plotting all the inequalities corresponding to the constraints on an XY graph.
- The common region formed by the intersection of these plotted inequalities called as **feasible region (optimal region/ convex region)**.
- To determine the optimum solution, we calculate the objective function's value at each corner point within this feasible region.
- By comparing these values, we can identify and establish the optimal solution for the given LPP, because
- **“Extreme values occur at the vertices of a concave, bounded, feasible region”**

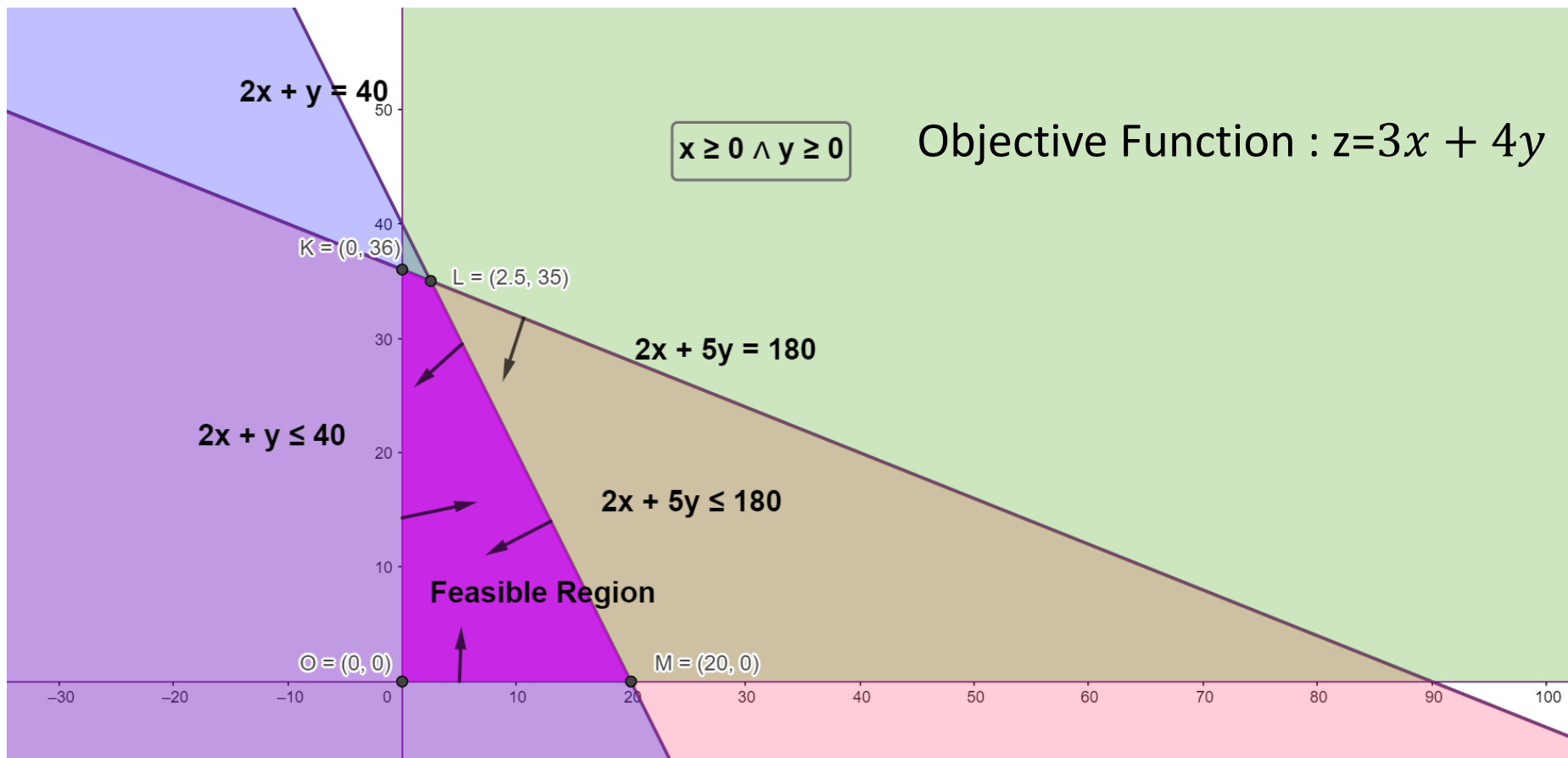
- **Problem 01:**

Maximize: $3x + 4y$

Subjected to constraints: $4x + 2y \leq 80$, $2x + 5y \leq 180$, $x, y \geq 0$ using graphical method.

Solution:

Given constraints	$4x + 2y \leq 80$ $\Rightarrow 2x + y \leq 40$	$2x + 5y \leq 180$
Replace inequality by equality	$\Rightarrow 2x + y = 40$	$2x + 5y = 180$
Find the x -intercept by taking $y = 0$ y -intercept by taking $x = 0$	$y = 0 \Rightarrow x = 10$ $x = 0 \Rightarrow y = 40$	$y = 0 \Rightarrow x = 90$ $x = 0 \Rightarrow y = 36$
Draw the lines joining points	$(10,0) \& (0,40)$	$(90,0) \& (0,36)$
Shade the region of inequality by checking $(0,0)$ lies within the region. i.e., put $(0,0)$ into the inequality; if it satisfies the inequality then the region will be toward the origin else it will be on other side	$4(0) + 2(0) \leq 80$ $\Rightarrow 0 \leq 80$ Satisfied Hence mark toward $(0,0)$	$2(0) + 5(0) \leq 180$ $\Rightarrow 0 \leq 180$ Satisfied Hence mark toward $(0,0)$



Calculate z value at each vertices and find the minimum and maximum value of z

$K(0,36)$	$z(0,36) = 144$	
$L(2.5,3.5)$	$z(2.5,35) = 147.5$	z is maximum
$M(20,0)$	$z(20,0) = 60$	
$O(0,0)$	$z(0,0) = 0$	z is minimum

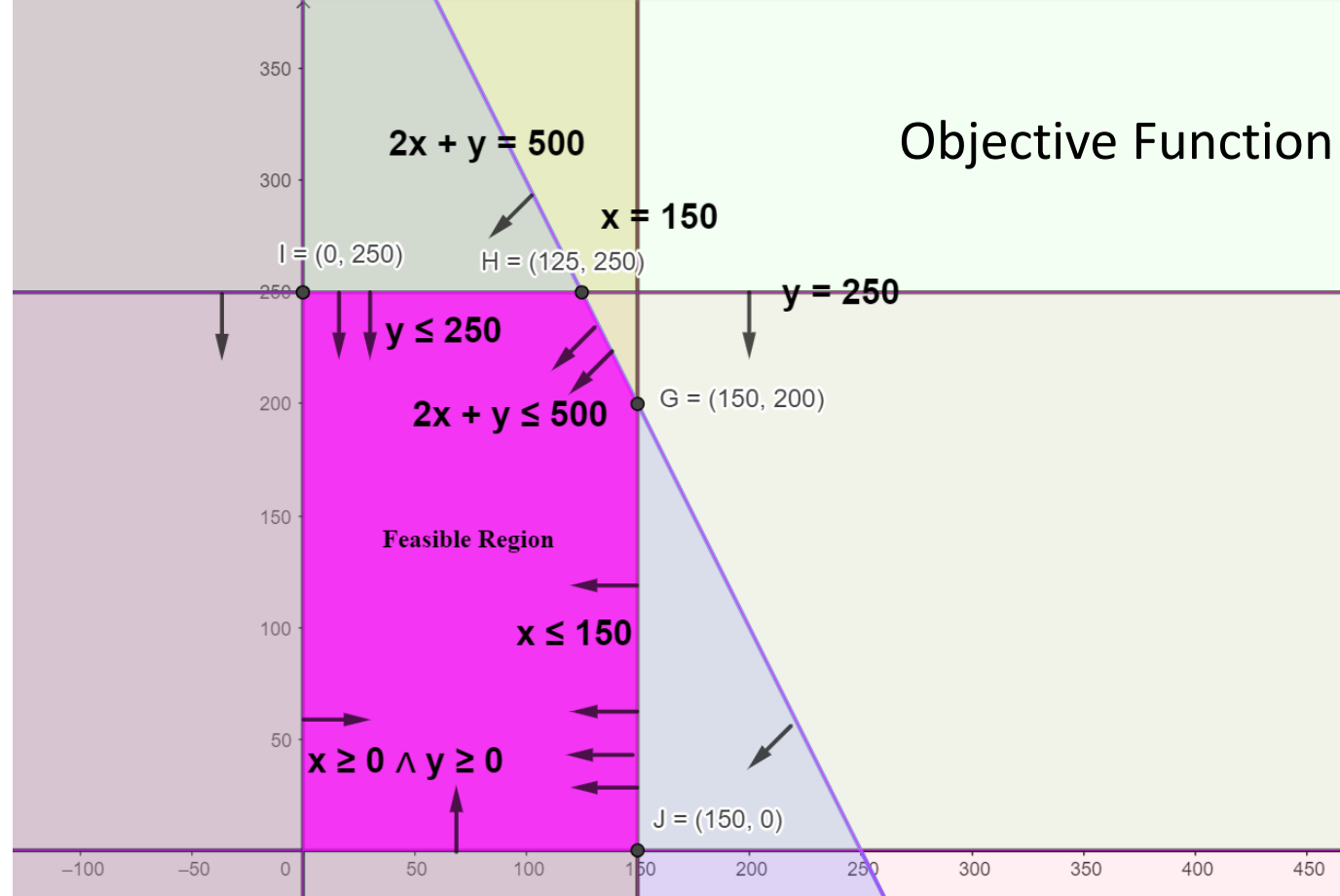
- **Problem 02:**

Maximize: $8x + 5y$

Subjected to constraints: $2x + y \leq 500, x \leq 150, y \leq 250, x, y \geq 0$ using graphical method.

Solution:

Given constraints	$2x + y \leq 500$	$x \leq 150$	$y \leq 250$
Replace inequality by equality	$2x + y = 500$	$x = 150$	$y = 250$
Find the x -intercept by taking $y = 0$ y -intercept by taking $x = 0$	$y = 0 \Rightarrow x = 500$ $x = 0 \Rightarrow y = 500$	$y = 0 \Rightarrow x = 250$ No y -intercept	$x = 0 \Rightarrow y = 250$ No x -intercept
Draw the lines joining points	$(250,0) \& (0,500)$	$(250,0)$	$(0,250)$
Shade the region of inequality by checking $(0,0)$ lies within the region. i.e., put $(0,0)$ into the inequality; if it satisfies the inequality then the region will be toward the origin else it will be on other side	$2(0) + (0) \leq 500$ $\Rightarrow 0 \leq 500$ Satisfied Hence mark toward $(0,0)$	$(0) \leq 150$ Satisfied Hence mark toward $(0,0)$	$(0) \leq 250$ Satisfied Hence mark toward $(0,0)$



Calculate z value at each vertices and find the minimum and maximum value of z

$I(0,250)$	$z(0,250) = 1250$	
$H(125,250)$	$z(125,250) = 2250$	z is maximum
$G(150,200)$	$z(150,200) = 2200$	
$J(150,0)$	$z(150,0) = 1200$	

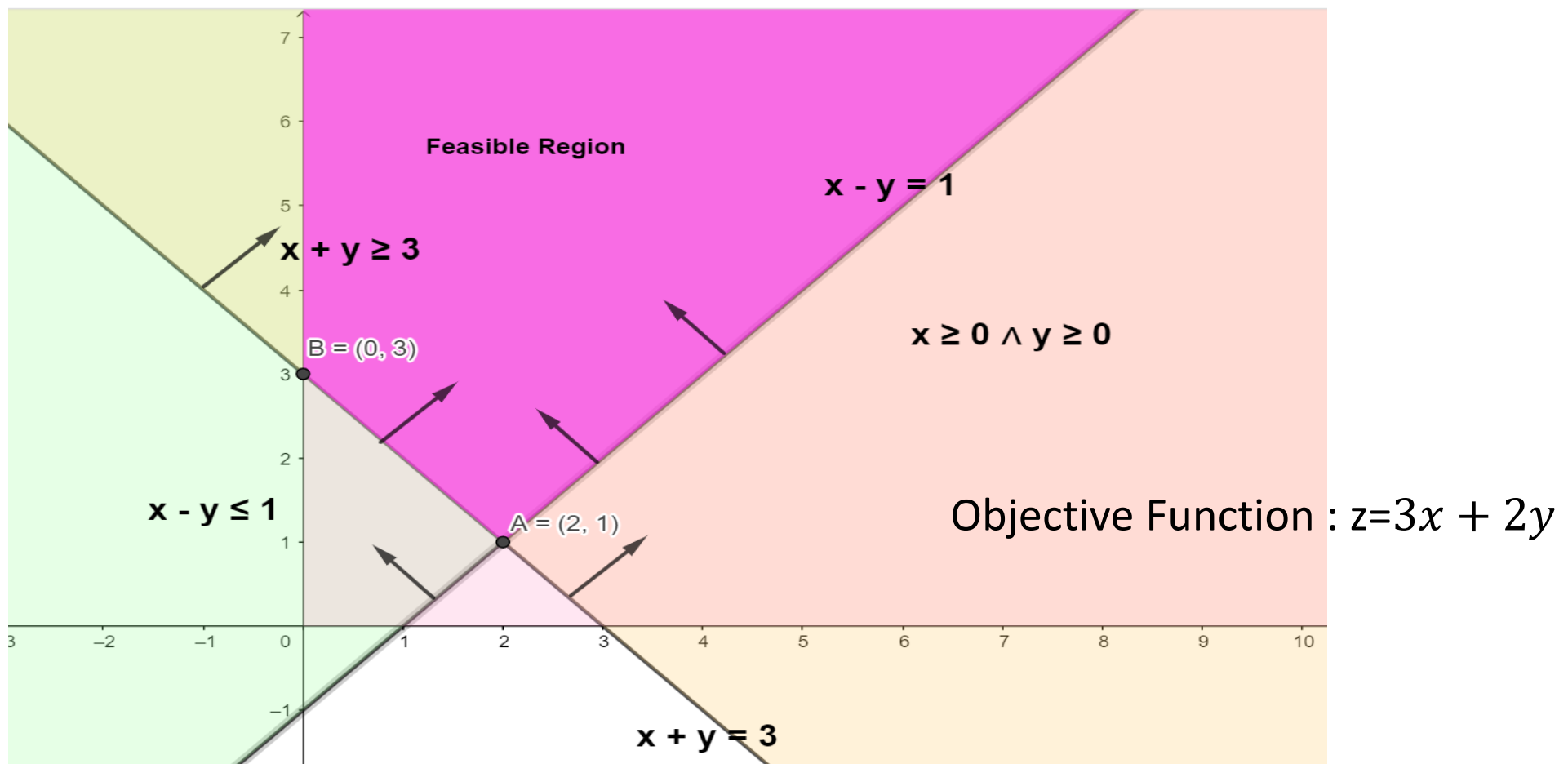
- **Problem 03:**

Minimize: $3x + 2y$

Subjected to constraints: $x - y \leq 1, x + y \geq 3, x, y \geq 0$ using graphical method.

Solution:

Given constraints	$x - y \leq 1$	$x + y \geq 3$
Replace inequality by equality	$x - y = 1$	$x + y = 3$
Find the x -intercept by taking $y = 0$ y -intercept by taking $x = 0$	$y = 0 \Rightarrow x = 1$ $x = 0 \Rightarrow y = -1$	$y = 0 \Rightarrow x = 3$ $x = 0 \Rightarrow y = 3$
Draw the lines joining points	$(1,0) \& (0,-1)$	$(3,0) \& (0,3)$
Shade the region of inequality by checking $(0,0)$ lies within the region. i.e., put $(0,0)$ into the inequality; if it satisfies the inequality then the region will be toward the origin else it will be on other side	$(0) - (0) \leq 1$ $\Rightarrow 0 \leq 1$ Satisfied Hence mark toward $(0,0)$	$(0) + (0) \geq 3$ Not satisfied Hence mark other side of $(0,0)$



Calculate z value at each vertices and find the minimum and maximum value of z

$A(2,1)$	$z(2,1) = 8$	
$H(0,3)$	$z(0,3) = 6$	z is minimum

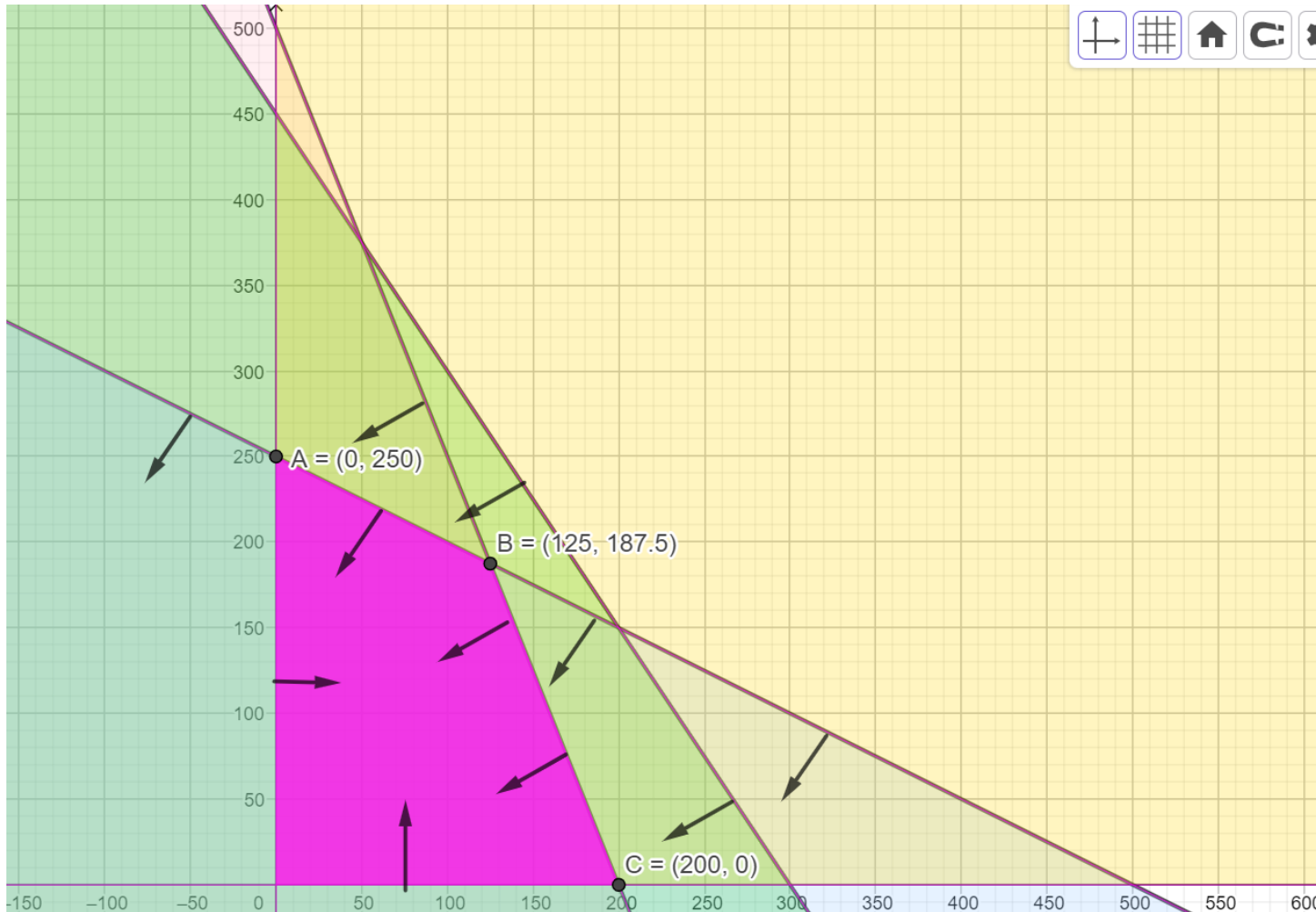
- If this problem is of Maximization, since the feasible region is unbounded, no solution for the problem Max z

- **Problem 04:**

Minimize: $100x + 40y$

Subjected to constraints: $5x + 2y \leq 1000$, $3x + 2y \leq 900$, $x + 2y \leq 500$, $x, y \geq 0$ using graphical method.

Solution:



$A(0, 250)$	$z(0, 250) = 10000$
$B(125, 187.5)$	$z(125, 187.5) = 20000$
$C(200, 0)$	$z(200, 0) = 20000$

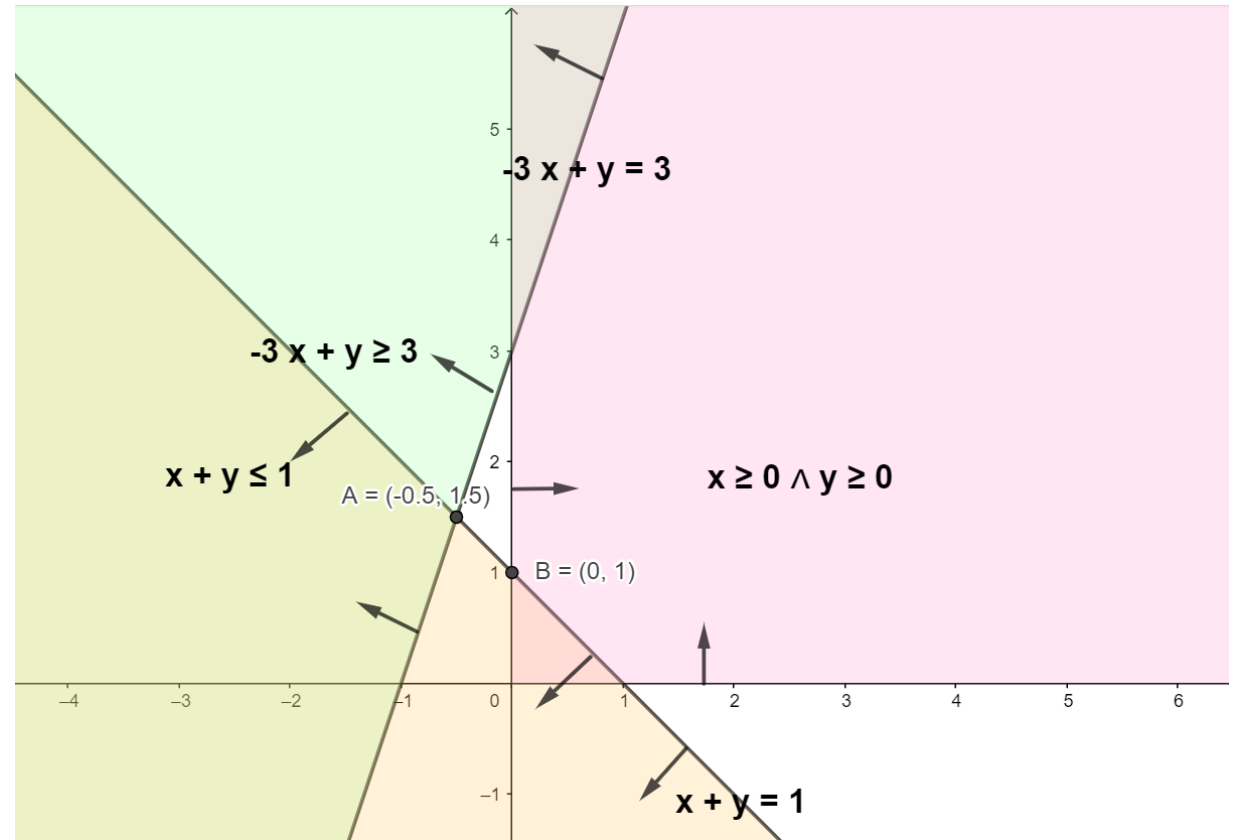
Since z is maximum at two points B and C , optimal solution exists at all the points on the line joining points B and C .

- **Problem 05:**

Maximize: $x + y$

Subjected to constraints: $x + y \leq 1$, $-3x + y \geq 3$, $x, y \geq 0$ using graphical method.

Solution: Since there is no common region which satisfies all the constraint, given LPP has no optimal solution.



A manufacturing company makes two types of television sets; one is black and white and the other is color. The company has resources to make at most 300 sets a week. It takes Rs 1800 to make a black and white set and Rs 2700 to make a coloured set. The company can spend no more than Rs 648000 a week to make television sets. If it makes a profit of Rs 510 per black and white set and Rs 675 per coloured set, how many sets of each type should be produced so that the company has a maximum profit? Formulate this problem as a LPP given that the objective is to maximize the profit

Hint: x – Number of black and white TVs

y – Number of color TVs

The company has resources to make at most 300 sets a week.

$$\Rightarrow x + y \leq 300$$

It takes Rs 1800 to make a black and white set and Rs 2700 to make a coloured set.

$\Rightarrow 1800x + 2700y \rightarrow$ Cost to make both black and white TVs and colored TVs

The company can spend no more than Rs 648000 a week to make television sets.

$$\Rightarrow 1800x + 2700y \leq 648000$$

If it makes a profit of Rs 510 per black and white set and Rs 675 per coloured set,

Profit = $z = 510x + 675y$ is to be maximized

$\Rightarrow$	Max	$:z = 510x + 675y$
Subjected to		$:x + y \leq 300$
		$1800x + 2700y \leq 648000$
		$x, y \geq 0$

Solve the above LPP using graphical method

- Any LPP can have
 - A unique optimal solution (Problem 01)
 - An unbounded solutions(Problem 03)
 - An infinite number of solutions (Problem 04)
 - No solution (Problem 05)
- LPP can be classified as follows:
- **Standard form of LPP:**
 - Objective function should be of type Maximization
 - All constraints are expressed as equations
 - RHS of the each constraint should be non-negative
 - All variables are non-negative
- **Canonical form of LPP**
 - Objective function should be of type Maximization
 - All constraints are of $\leq$ type except non-negative constraints
 - All variables are non-negative

Note:

- If the problem is not in canonical or standard form then we can bring the given LPP into these forms by performing following steps:
 - Minimize z can be taken as $-\text{Maximize } (-z)$
 - Example: Minimize $z = x - 2y + 3z \approx -\text{Maximize } (-z) = z^* = -x + 2y - 3z$
 - Inequalities can be converted to equalities by introducing new variables called as **slack variable** and **surplus variable**
 - The variables added to the constraints of type $\leq$ are called as **slack variable**
 - Example: $x + y \leq 20 \rightarrow x + y + s_1 = 20$, here s_1 is the slack variable
 - The variables subtracted to the constraints of type $\geq$ are called as **surplus variable**
 - Example: $x + y \geq 20 \rightarrow x + y - s_1 = 20$, here s_1 is the surplus variable

Simplex method

- It is not possible to solve multivariable LPP using graphical method, hence we have a new method called as **Simplex method**.
- Simplex method is an iterative algorithm to find the optimal value of the multivariable objective function

Steps to solve LPP using simplex method:

- **Standard form:** Convert given LPP into standard form

LPP	Std. LPP
Min: $z = z_1x_1 - z_2x_2 + \dots$	$-Max(-z) = z^* = -z_1x_1 + z_2x_2 - \dots$

- **Introducing slack variables:** Determine the slack variable and convert given inequalities into equalities

LPP	Std. LPP
$\begin{aligned} z &= -z_1x_1 + z_2x_2 - \dots \\ a_1x_1 + a_2x_2 + \dots &\leq b_1 \\ &\vdots \end{aligned}$	$\begin{aligned} z &= -z_1x_1 + z_2x_2 + \dots - 0s_1 - 0s_2 - \dots \\ a_1x_1 + a_2x_2 + \dots + s_1 + 0s_2 + \dots &= b_1 \\ &\vdots \end{aligned}$

- **Creating the table:** Construct a simplex table to perform row operations on the linear programming model as well as to check a solution for optimality. The table consists of the coefficient corresponding to the linear constraint variables and the coefficients of the objective function in the bottom row.
- **Basis:** Given a system of m linear equations with n variables ($m < n$) any solutions which is obtained by solving m variables keeping $n - m$ variables 0 called as basic solution and such m variables are called as basic variables (basis) and $(n - m)$ variables are called as non-basic variables.
- **Simplex algorithm is an iterative process which improves the basic solution**

Basis	x_1	x_2	x_3	...	s_1	s_2	...	RHS
s_1	a_1	a_2	a_3	...	1	0	...	b_1
s_2	c_1	c_2	c_3	...	0	1	...	b_1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
z	z_1	z_2	z_3	...	0	0	...	0

- **Pivot variables:** Pick the smallest negative value in the bottom row. One of the values lying in the column (Key column) of this value will be the pivot variable (Key element). To find the indicator (Key row), divide the RHS values of the linear constraints by their corresponding values from the column containing the possible key element.

Basic Variables
-simply which are
added to the
constraints

Basis	x_1	x_2	s_1	s_2	RHS	RHS/Key column
s_1	2	1	1	0	3	3/1
s_2	3	5	0	1	9	9/5
z	-1	-4	0	0	0	0

1. Smallest negative value
2. Key column
3. Smallest positive ratio
4. Key row
5. Key element

Note: Columns corresponding to Basic variables should form an identity matrix.

- **Creating a new tableau**
 - To optimize the key element, it will need to be transformed into a unit value (value of 1). To transform the value, multiply the row containing the key element by the reciprocal of the key element.
 - After the unit value has been determined, the other values in the column containing the unit value will become zero. The basic variable in the key row replaced with the variable in the key column called as **outgoing and incoming variables**.
- **Checking for optimality**
 - An optimal solution is defined as one where all values in the bottom row are greater than or equal to zero. If every value meets this criterion, the solution is deemed optimal.

Problem 01: Solve the following LPP using Simplex method:

$$\text{Max } z = 3x_1 + 2x_2$$

$$\text{S.T.C. : } x_1 + x_2 \leq 4$$

$$x_1 - x_2 \leq 2, \quad x_1, x_2 \geq 0$$

- Convert it to standard form of LPP
- Objective function should be of type Maximization

$$\text{Max } z = 3x_1 + 2x_2$$

- Add slack variables to constraints to convert $\leq \rightarrow =$

$$\text{S.T.C. } x_1 + x_2 + s_1 = 4$$

$$x_1 - x_2 + s_2 = 2$$

- Also introduce slack variables in objective function, hence the standard form of LPP will be

$$\text{Max } z = 3x_1 + 2x_2 - 0s_1 - 0s_2$$

$$\text{S.T.C. } x_1 + x_2 + s_1 = 4$$

$$x_1 - x_2 + s_2 = 2$$

$$x_1, x_2, s_1, s_2 \geq 0$$

- Construct the table

Basis	x_1	x_2	s_1	s_2	<i>RHS</i>	Ratio
s_1	1	1	1	0	4	4/1=4
s_2	1	-1	0	1	2	2/1=2
Obj. func.	-3	-2	0	0	0	

- -3 is the smallest negative number in the last row
- Column corresponding to the number -3 is the key column
- Find the ratio of RHS/Key column
- 2 is the smallest positive ratio
- Row corresponding to the number 2 is the key row
- Intersection of key row and key column is the key element, here 1 is the key element.
- Leaving variable will be s_2 and entry variable will be x_1 , i.e., replace s_2 in the first column by x_1
- Now corresponding columns s_1 and x_1 should form the identity matrix, hence
- If key element is not equal to 1 then make the key element 1 by dividing entire key row by that element
- Perform row operations to make key element 1 and make all numbers above and below the key element become 0.
 - Here perform $R'_1 \rightarrow R_1 - R_2$ and $R'_3 \rightarrow R_3 + 3R_2$

s_1	$\rightarrow$	s_1
s_2		x_1

Basis	x_1	x_2	s_1	s_2	<i>RHS</i>	Ratio
s_1	0	2	1	-1	2	1
x_1	1	-1	0	1	2	-2
Obj. func.	0	-5	0	3	6	

Basis	x_1	x_2	s_1	s_2	<i>RHS</i>	Ratio
s_1	0	2	1	-1	2	1
x_1	1	-1	0	1	2	-2
Obj. func.	0	-5	0	3	6	

- -5 is the smallest negative number in the last row
- Column corresponding to the number -5 is the key column
- Find the ratio of RHS/Key column
- 1 is the smallest positive ratio (Don't consider negative values)
- Row corresponding to the number 1 is the key row
- Intersection of key row and key column is the key element, here 2 is the key element.
- Leaving variable will be s_1 and entry variable will be x_2 , i.e., replace s_1 in the first column by x_2
- Now corresponding columns x_2 and x_1 should form the identity matrix, hence
- The key element is not equal to 1 then make the key element 1 by dividing entire key row by that element
- Perform row operations to make key element 1 and all numbers above and below the key element become 0.
 - Here perform $R'_1 \rightarrow R_1/2$; $R'_2 \rightarrow R_2 + R_1$; $R'_3 \rightarrow R_3 + 5R_1$

s_1		x_2
x_1		x_1

Basis	x_1	x_2	s_1	s_2	<i>RHS</i>	Ratio
x_2	0	1	1/2	-1/2	1	
x_1	1	0	1/2	1/2	3	
Obj. func.	0	0	5/2	1/2	11	

There will be no negative number in the last row, hence we obtained the optimal solution. $x_1 = 3, x_2 = 1$ and $z = 11$

Problem 02: Solve the following LPP using Simplex method:

$$\text{Min } u = x - 3y - 2z$$

$$\text{S.T.C. : } 3x - y + 2z \leq 7$$

$$-2x + 4y \leq 12,$$

$$-4x + 3y + 8z \leq 10, x, y, z \geq 0$$

➤ Convert it to standard form of LPP

➤ Objective function should be of type Maximization

$$\text{Min } u = -\text{Max}(u^*), u^* = -u = -x + 3y + 2z$$

➤ Add slack variables to constraints to convert $\leq \rightarrow =$

$$\text{S.T.C. } 3x - y + 2z + s_1 = 7$$

$$-2x + 4y + s_2 = 12$$

$$-4x + 3y + 8z + s_3 = 10$$

➤ Also introduce slack variables in objective function, hence the standard form of LPP will be

$$\text{Max } u^* = -x + 3y + 2z - 0s_1 - 0s_2 - 0s_3$$

$$\text{S.T.C. } 3x - y + 2z + s_1 = 7$$

$$-2x + 4y + s_2 = 12$$

$$-4x + 3y + 8z + s_3 = 10$$

$$x, y, z, s_1, s_2, s_3 \geq 0$$

➤ Construct the table

Basis	x	y	z	s_1	s_2	s_3	<i>RHS</i>	Ratio
s_1	3	-1	2	1	0	0	7	-7
s_2	-2	4	0	0	1	0	12	3
s_3	-4	3	8	0	0	1	10	10/3
Obj. func.	1	-3	-2	0	0	0	0	

- -3 is the smallest negative number in the last row
- Column corresponding to the number -3 is the key column
- Find the ratio of RHS/Key column
- 3 is the smallest positive ratio (Don't consider negative values)
- Row corresponding to the number 3 is the key row
- Intersection of key row and key column is the key element, here 4 is the key element.
- Leaving variable will be s_2 and entry variable will be y , i.e., replace s_2 in the first column by y 
- Now corresponding columns s_1 y and s_3 should form the identity matrix, hence
- If the key element 4 (not equal to 1) then make the key element 1 by dividing entire key row by 4
- Perform row operations to make key element 1 and all numbers above and below the key element become 0.
 - Here perform $R'_2 \rightarrow R_2/4$; $R'_1 \rightarrow R_1 + R_2$; $R'_3 \rightarrow R_3 - R_2$

s_1		s_1
s_2		y
s_3		s_3

Basis	x	y	z	s_1	s_2	s_3	<i>RHS</i>	Ratio
s_1	5/2	0	2	1	1/4	0	10	5
y	-1/2	1	0	0	1/4	0	3	-
s_3	-5/2	0	8	0	-3/4	1	1	1/8
Obj. func.	-1/2	0	-2	0	3/4	0	9	

Basis	x	y	z	s_1	s_2	s_3	<i>RHS</i>	Ratio
s_1	5/2	0	2	1	1/4	0	10	5
y	-1/2	1	0	0	1/4	0	3	-
s_3	-5/2	0	8	0	-3/4	1	1	1/8
Obj. func.	-1/2	0	-2	0	3/4	0	9	

- -2 is the smallest negative number in the last row
- Column corresponding to the number -2 is the key column
- Find the ratio of RHS/Key column
- $1/8$ is the smallest positive ratio (Don't consider negative values)
- Row corresponding to the number $1/8$ is the key row
- Intersection of key row and key column is the key element, here 8 is the key element.
- Leaving variable will be s_3 and entry variable will be z , i.e., replace s_3 in the first column by z 
- Now corresponding columns s_1 , y and z should form the identity matrix, hence
- If key element 8 is not equal to 1 then make the key element 1 by dividing entire key row by 8
- Perform row operations to make key element 1 and all numbers above and below the key element become 0.
 - Here perform $R'_3 \rightarrow R_3/8$; $R'_1 \rightarrow R_1 - 2R_3$; $R'_2 \rightarrow R_2$; $R'_4 \rightarrow R_4 + 2R_3$

s_1		s_1
y		x
s_3		z

Basis	x	y	z	s_1	s_2	s_3	<i>RHS</i>	Ratio
s_1	25/8	0	0	1	7/16	-1/4	39/4	
y	-1/2	1	0	0	1/4	0	3	
z	-5/16	0	1	0	-3/32	1/8	1/8	
Obj. func.	-9/8	0	0	0	9/16	1/4	9.25	

Basis	x	y	z	s_1	s_2	s_3	<i>RHS</i>	Ratio
s_1	25/8	0	0	1	7/16	-1/4	39/4=9.75	3.12
y	-1/2	1	0	0	1/4	0	3	-
z	-5/16	0	1	0	-3/32	1/8	1/8=0.125	-
Obj. func.	-9/8	0	0	0	9/16	1/4	9.25	

- $-9/8$ is the smallest negative number in the last row
- Column corresponding to the number $-9/8$ is the key column
- Find the ratio of RHS/Key column
- 3.12 is the smallest positive ratio (Don't consider negative values)
- Row corresponding to the number 3.12 is the key row
- Intersection of key row and key column is the key element, here 25/8 is the key element.
- Leaving variable will be s_1 and entry variable will be x , i.e., replace s_1 in the first column by x 
- Now corresponding columns x, y and z should form the identity matrix, hence
- If key element is not equal to 1 then make the key element 1 by dividing entire key row by 4
- Perform row operations to make key element 1 and all numbers above and below the key element become 0.
 - Here perform $R'_1 \rightarrow R_1 \cdot 8/25$; $R'_2 \rightarrow R_2 + R_1/2$; $R'_3 \rightarrow R_3 + 5R_2/16$; $R'_4 \rightarrow R_4 + 9R_1/8$

s_1		x
y		y
z		z

Basis	x	y	z	s_1	s_2	s_3	<i>RHS</i>	Ratio
x	1	0	0	0.32	0.14	-0.08	3.12	
y	0	1	0	0.16	0.32	-0.04	4.56	
z	0	0	1	0.1	-0.05	0.1	1.1	
Obj. func.	0	0	0	0.36	0.72	0.16	12.76	

There will be no negative number in the last row, hence we obtained the optimal solution.

$$\begin{aligned}
 x &= 3.12, y = 4.56 \\
 z &= 1.1 \text{ and } u^* = 12.76 \\
 \Rightarrow \text{Min } u &= -12.76
 \end{aligned}$$

Problem 03: Solve the following LPP using Simplex method:

Min $z = x + y$

S.T.C. : $x + y \leq 1$

$-3x + y \geq 3, \quad x, y \geq 0$

- Convert it to standard form of LPP
- Objective function should be of type Maximization
 $\text{Min } z = -\text{Max}(z^*), z^* = -z = -x - y$
- Convert $\geq$ to $\leq$ in constraints by multiplying $-$ sign on both side except non-negative constraints

S.T.C. : $x + y \leq 1$

$3x - y \leq -3, \quad x, y \geq 0$

- Add slack variables to constraints to convert $\leq \rightarrow =$

S.T.C. $x + y + s_1 = 1$

$3x - y + s_2 = -3, \quad x, y \geq 0$

- Also introduce slack variables in objective function, hence the standard form of LPP will be

$\text{Max } z^* = -x - y - 0s_1 - 0s_2$

S.T.C. $x + y + s_1 = 1$

$3x - y + s_2 = -3, \quad x, y, s_1, s_2 \geq 0$

- Construct the table

Basis	x	y	s_1	s_2	<i>RHS</i>	Ratio
s_1	1	1	1	0	1	
s_2	3	-3	0	1	-3	
Obj. func.	1	1	0	0	0	

There are no negative number in the last row, hence we arrived at the optimal level.

Since x and y are not present in the first column, we shall take $x = 0, y = 0$.

But this is not the optimal solution, because this violates the condition that $-3x + y \geq 3$.

Hence, the given LPP has no solution.

Note: In the Simplex method, if there are multiple smallest negative (or smallest positive) ratios, you are free to select any one of them.

Big-M Method

- LPP should be of type Maximization
- Introduce slack variables to the $\leq$ constraints
- Introduce artificial variables along with the surplus variables to $\geq$ constraints
- Introduce only artificial variables to constraint equations
- Addition of these artificial variables causes violation of the corresponding constraints, hence assign a very large penalty ($-M$ for maximization problem) to the objective function.
- Use simplex method to solve the obtained LPP.
- **Note:**
 1. If an **artificial variable remains** in the first column at the optimal level of the simplex table, then the Linear Programming Problem (LPP) is unsolvable.
 2. Whenever an artificial variable happens to leave in the first column, remove the artificial variable column

Problem 01: Solve LP problem using big M method (Penalty method).

$$\text{Max } Z = 3x_1 - x_2$$

Subjected to

$$2x_1 + x_2 \geq 2$$
$$x_1 + 3x_2 \leq 3$$
$$x_2 \leq 4, \quad x_1, x_2 \geq 0$$

Step1 : Introduce slack, surplus and artificial variables also penalty M in the objective function

$$\text{Max } Z = 3x_1 - x_2 - 0s_1 - 0s_2 - 0s_3 - MA_1$$

Subjected to

$$2x_1 + x_2 - s_1 + A_1 = 2$$
$$x_1 + 3x_2 + s_2 = 3$$
$$x_2 + s_3 = 4$$
$$x_1, x_2, s_1, s_2, s_3, A_1 \geq 0$$

Basis	x_1	x_2	s_1	s_2	s_3	A_1	RHS	Ratio
A_1	2	1	-1	0	0	1	2	
s_2	1	3	0	1	0	0	3	
s_3	0	1	0	0	1	0	4	
Obj. Func	-3	1	0	0	0	M	0	

Note:

As the columns corresponding to the basic variables A_1, s_2 and s_3 do not conform to the standard form, it implies that other than the intersection of the A_1 (and s_2, s_3) row with the A_1 (and s_2, s_3) column, all entries above and below should be zero. Here we can see M in the last row. Eliminate M from row operation

Step2: Eliminate M in the objective function: $R'_4 \rightarrow R_4 - MR_1$

Step3: Construct the table

Basis	x_1	x_2	s_1	s_2	s_3	A_1	RHS	Ratio
A_1	2	1	-1	0	0	1	2	1
s_2	1	3	0	1	0	0	3	3
s_3	0	1	0	0	1	0	4	-
Obj. Func	$-(3 + 2M)$	$-(M - 1)$	M	0	0	0	$-2M$	

Step4: Apply simplex method to get the optimal solution

Note: Since the value of the penalty M is very big, we can assume M value as 10(or 100, 1000 also) for our convience. Then $-(3 + 2M) = -23$ and $-(M - 1) = -9$. $-23 < -9$ hence $-(3 + 2M)$ is the smallest negative number in the last row

Basis	x_1	x_2	s_1	s_2	s_3	A_1	RHS	Ratio
A_1	2	1	-1	0	0	1	2	1
s_2	1	3	0	1	0	0	3	3
s_3	0	1	0	0	1	0	4	-
Obj. Func	$-(3 + 2M)$	$-(M - 1)$	M	0	0	0	$-2M$	

- $-(3 + 2M)$ is the smallest negative number in the last row
- Column corresponding to the number $-(3 + 2M)$ is the key column
- Find the ratio of RHS/Key column
- 1 is the smallest positive ratio (Don't consider negative values)
- Row corresponding to the number 1 is the key row
- Intersection of key row and key column is the key element, here 2 is the key element.
- Leaving variable will be A_1 and entry variable will be x_1 , i.e., replace A_1 in the first column by x_1
- Now corresponding columns A_1, s_2 and s_3 should form the identity matrix, hence
- The key element is 2 (not equal to 1 then make the key element 1 by dividing entire key row by 2)
- Perform row operations to make all numbers above and below the key element become 0. (**Note: Remove A_1 column as A_1 will be removed from the first column, this is applicable only for artificial variables**)
 - Here perform $R'_1 \rightarrow R_1/2$; $R'_2 \rightarrow R_2 - R_1$; $R'_3 \rightarrow R_3$; $R'_4 \rightarrow R_4 + (3 + 2M)R_1$

Basis	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
x_1	1	0.5	-0.5	0	0	1	
s_2	0	2.5	0.5	1	0	2	
s_3	0	1	0	0	1	4	
Obj. Func	0	2.5	-1.5	0	0	3	

Basis	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
x_1	1	0.5	-0.5	0	0	1	-2
s_2	0	2.5	0.5	1	0	2	4
s_3	0	1	0	0	1	4	
Obj. Func.	0	2.5	-1.5	0	0	3	

- -1.5 is the smallest negative number in the last row
- Column corresponding to the number -1.5 is the key column
- Find the ratio of RHS/Key column
- 4 is the smallest positive ratio (Don't consider negative values)
- Row corresponding to the number 4 is the key row
- Intersection of key row and key column is the key element, here 0.5 is the key element.
- Leaving variable will be s_2 and entry variable will be s_1 , i.e., replace s_1 in the first column by s_1
- Now corresponding columns x_1, s_1 and s_3 should form the identity matrix, hence
- The key element is 0.5 (not equal to 1 then make the key element 1 by dividing entire key row by 0.5 or multiply 2)
- Perform row operations to make all numbers above and below the key element become 0.
 - Here perform $R'_2 \rightarrow 2R_2$; $R'_1 \rightarrow R_1 + R_2/2$; $R'_3 \rightarrow R_3$; $R'_4 \rightarrow R_4 + 3R_1/2$

Basis	x_1	x_2	s_1	s_2	s_3	RHS
x_1	1	3	0	1	0	3
s_1	0	5	1	2	0	4
s_3	0	1	0	0	1	4
Obj. Func.	0	10	0	3	0	9

Hence, no negative number in the last row, hence we obtained the optimal solution.
 $x_1 = 3, x_2 = 0$ and maximum value of $Z = 9$

Problem 02: Solve LP problem using big M method (Penalty method).

$$\text{Max } Z = 3x_1 + 2x_2$$

$$\begin{aligned} \text{Subjected to } & 2x_1 + x_2 \leq 2 \\ & 3x_1 + 4x_2 \geq 12 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Step1 : Introduce slack, surplus and artificial variables also penalty M in the objective function

$$\text{Max } Z = 3x_1 + 2x_2 - 0s_1 - 0s_2 - MA_1 \quad -(1)$$

$$\text{Subjected to } 2x_1 + x_2 + s_1 = 2 \quad -(2)$$

$$3x_1 + 4x_2 - s_2 + A_1 = 12 \quad -(3)$$

$$x_1, x_2, s_1, s_2, A_1 \geq 0$$

Step2: Eliminate M in the objective function: $(1) + M(3)$

Basis	x_1	x_2	s_1	s_2	A_1	RHS	Ratio
s_1	2	1	1	0	0	2	
A_1	3	4	0	-1	1	12	
Obj. Func	-3	-2	0	0	M	0	

Note:

As the columns corresponding to the basic variables s_1 and A_1 do not conform to the standard form, it implies that other than the intersection of the s_1 (and A_1) row with the s_1 (and A_1) column, all entries above and below should be zero. Here we can see M in the last row. Eliminate M by row operation

Step2: Eliminate M in the objective function: $R'_3 \rightarrow R_3 - MR_2$

Step3: Construct the table

Basis	x_1	x_2	s_1	s_2	A_1	RHS	Ratio
s_1	2	1	1	0	0	2	
A_1	3	4	0	-1	1	12	
Obj. Func	$-(3 + 3M)$	$-(2 + 4M)$	0	M	0	$-12M$	

Step4: Apply simplex method to get the optimal solution

Note: Since the value of the penalty M is very big, we can assume M value as 10(or 100, 1000 also) for our convience. Then $-(3 + 3M) = -33$ and $-(2 + 4M) = -42$. $-42 < -33$ hence $-(2 + 4M)$ is the smallest negative number in the last row

Basis	x_1	x_2	s_1	s_2	A_1	RHS	Ratio
s_1	2	1	1	0	0	2	2
A_1	3	4	0	-1	1	12	3
Obj. Func	$-(3 + 3M)$	$-(2 + 4M)$	0	M	0	$-12M$	

- $-(2 + 4M)$ is the smallest negative number in the last row
- Column corresponding to the number $-(2 + 4M)$ is the key column
- Find the ratio of RHS/Key column
- 2 is the smallest positive ratio (Don't consider negative values)
- Row corresponding to the number 2 is the key row
- Intersection of key row and key column is the key element, here 1 is the key element.
- Leaving variable will be s_1 and entry variable will be x_2 , i.e., replace s_1 in the first column by x_2
- Now corresponding columns x_2 and A_1 should form the identity matrix, hence
- The key element is 1 (if not equal to 1 then make the key element 1 by dividing entire key row by that key element)
- Perform row operations to make all numbers above and below the key element become 0.
 - Here perform $R'_2 \rightarrow R_2 - 4R_1$; $R'_3 \rightarrow R_3 + (2 + 4M)R_1$

Basis	x_1	x_2	s_1	s_2	A_1	RHS	Ratio
x_2	2	1	1	0	0	2	
A_1	-5	0	-4	-1	1	4	
Obj. Func	$5M + 1$	0	$4M + 2$	M	0	$4 - 4M$	

For some large value of M say 10, $5M + 1 = 51 > 0$, $4M + 2 = 42 > 0$ and $M = 10 > 0$. Hence, no negative number in the last row, hence we obtained the optimal level. **But A_1 is present in the first column, hence NO SOLUTION**

Problem 03: Solve LP problem using big M method (Penalty method).

$$\text{Min } Z = 2x_1 + 3x_2$$

$$\begin{aligned} \text{Subjected to } & x_1 + x_2 \geq 5 \\ & x_1 + 2x_2 \geq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Step1 : Convert Min Z in to Maximization problem, then introduce slack, surplus and artificial variables also penalty M in the objective function

$$\text{Max } Z^* = -z = -2x_1 - 3x_2 - 0s_1 - 0s_2 - MA_1 - MA_2 \quad -(1)$$

$$\text{Subjected to } x_1 + x_2 - s_1 + A_1 = 5 \quad -(2)$$

$$x_1 + 2x_2 - s_2 + A_2 = 6 \quad -(3)$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

Basis	x_1	x_2	s_1	s_2	A_1	A_2	RHS	Ratio
A_1	1	1	-1	0	1	0	5	5
A_2	1	2	0	-1	0	1	6	3
Obj. Func	2	3	0	0	M	M	0	

Step2: Eliminate M in the objective function: $R'_3 \rightarrow R_3 - MR_1 - MR_2$

Step3: Construct the table and follow the procedure of simplex method, we arrive at

Basis	x_1	x_2	s_1	s_2	A_1	A_2	RHS	Ratio
A_1	1	1	-1	0	1	0	5	5
A_2	1	2	0	-1	0	1	6	3
Obj. Func	$2 - 2M$	$3 - 3M$	M	M	0	0	$-11M$	

Basis	x_1	x_2	s_1	s_2	A_1	A_2	RHS	Ratio
A_1	1	1	-1	0	1	0	5	5
A_2	1	2	0	-1	0	1	6	3
Obj. Func	$2 - 2M$	$3 - 3M$	M	M	0	0	$-11M$	

Basis	x_1	x_2	s_1	s_2	A_1	RHS	Ratio
A_1	0.5	0	-1	0.5	1	2	4
x_2	0.5	1	0	-0.5	0	3	6
Obj. Func	$0.5(1-M)$	0	M	$0.5(3 - M)$	0	$-2M - 9$	

Basis	x_1	x_2	s_1	s_2	RHS	Ratio
x_1	1	0	-2	1	4	
x_2	0	1	1	-1	1	
Obj. Func	0	0	1	1	-11	

Hence, there are no negative numbers in the last row, optimal solution is $x_1 = 4, x_2 = 1$ and $Max Z^* = -11 \Rightarrow Min Z = -Max Z^* \Rightarrow 11$

2-phase Simplex Method

- The two-phase simplex method is another method to solve a given LPP involving some artificial variables. The solution is obtained in two phases

Phase 1

- In this phase, we construct an auxiliary LPP leading to a final simplex table containing a basic feasible solution to the original problem.

Step 1 Assign a cost -1 to each artificial variable and a cost 0 to all other variables and get a new objective function $Z^* = -A_1 - A_2 - A_3 \dots$ where A_i are artificial variables.

Step 2 Write down the auxiliary LPP in which the new objective function is to be maximized, subject to the given set of constraints.

Step 3 Solve the auxiliary LPP by simplex method until either of the following three cases arise:

- (i) $\text{Max } Z^* < 0$ and at least one artificial variable appears in the optimum basis at positive level.
- (ii) $\text{Max } Z^* = 0$ and at least one artificial variable appears in the optimum basis at zero level.
- (iii) $\text{Max } Z^* = 0$ and no artificial variable appears in the optimum basis.

In case (i), given LPP does not possess any feasible solution, whereas in cases (ii) and (iii) we go to phase II.

- Note: Whenever an artificial variable happens to leave in the first column, remove the artificial variable column

Phase II

Use the optimum basic feasible solution of phase I as a starting solution for the original LPP. Assign the actual costs to the variable in the objective function and a zero cost to every artificial variable in the basis at zero level. Delete the artificial variable column that is eliminated from the basis in phase I from the table. Apply simplex method to the modified simplex table obtained at the end of phase I till an optimum basic feasible solution is obtained or till there is an indication of unbounded solution.

Problem 01: Solve LP problem using two-phase simplex method

$$\text{Min } Z = -2x_1 - x_2$$

$$\text{Subjected to } x_1 + x_2 \geq 2$$

$$x_1 + x_2 \leq 4 \quad x_1, x_2 \geq 0$$

Write the given LPP in standard form

$$\text{Max } Z^* = 2x_1 + x_2 - 0s_1 - 0s_2 - A_1$$

$$\text{STC } x_1 + x_2 - s_1 + A_1 = 2$$

$$x_1 + x_2 + s_2 = 4, x_1, x_2, s_1, s_2 \geq 0$$

Phase 1 : Write the Auxiliary equation $\text{Max } Z^{**} = -A_1$

$$\text{STC } x_1 + x_2 - s_1 + A_1 = 2 \quad (1)$$

$$x_1 + x_2 + s_2 = 4, x_1, x_2, s_1, s_2 \geq 0$$

Basis	x_1	x_2	s_1	s_2	A_1	RHS	Ratio
A_1	1	1	-1	0	1	2	2
s_2	1	1	0	1	0	4	4
Obj. Func	0	0	0	0	1	0	

Columns of the basic variables does not form the identity matrix hence eliminate 1 using row operation: $R'_3 \rightarrow R_3 - R_1$

Now apply simplex algorithm to solve the obtained LPP

Basis	x_1	x_2	s_1	s_2	A_1	RHS	Ratio
A_1	1	1	-1	0	1	2	2
s_2	1	1	0	1	0	4	4
Obj. Func	-1	-1	1	0	0	-2	

Basis	x_1	x_2	s_1	s_2	RHS	Ratio
x_1	1	1	-1	0	2	
s_2	0	0	1	1	2	
Obj. Func	0	0	0	0	0	

Since there are no negative numbers in the last row, table is in optimum state. Hence continue to phase 2

Phase 2

Remove the column A_1 and replace the costs of the original objective function in the last row of the table obtained in phase 1

Basis	x_1	x_2	s_1	s_2	RHS	Ratio
x_1	1	1	-1	0	2	
s_2	0	0	1	1	2	
Obj. Func	-2	-1	0	0	0	

In Phase 2

1. Remove A_i 's columns
2. Keep all the rows except last row as it is and take the coefficients of the original obj function for the last row

Since x_1 and s_2 columns does not constitute the standard form, convert the table in to standard form, i.e., intersection of x_1 (and s_2) row with x_1 (and s_2) column =1 and all the elements above and below are 0 by performing row operation

$$R'_3 \rightarrow R_3 + 2R_1$$

Basis	x_1	x_2	s_1	s_2	RHS	Ratio
x_1	1	1	-1	0	2	2
s_2	0	0	1	1	2	-
Obj. Func	0	1	-2	0	4	

Here -2 is the smallest negative number

Basis	x_1	x_2	s_1	s_2	RHS	Ratio
x_1	1	1	-1	0	2	-
s_2	0	0	1	1	2	2
Obj. Func	0	1	-2	0	4	

Basis	x_1	x_2	s_1	s_2	RHS	Ratio
x_1	1	1	0	1	4	
s_1	0	0	1	1	2	-
Obj. Func	0	1	0	2	8	

Since there are no positive numbers in the last row, table is in optimum state. Hence the solution is

$$x_1 = 4, \quad x_2 = 0, \quad \text{Min } Z = -\text{Max } Z^* = -(8)$$

Problem 02: Solve LP problem using two-phase simplex method

$$\text{Max } Z = 5x_1 - 4x_2 + 3x_3$$

$$\text{Subjected to } 2x_1 + x_2 - 6x_3 = 20, \quad 6x_1 + 5x_2 + 10x_3 \leq 76, \quad 8x_1 - 3x_2 + 6x_3 \leq 50, \quad x_1, x_2, x_3 \geq 0$$

Write the given LPP in standard form

$$\text{Max } Z = 5x_1 - 4x_2 + 3x_3 - 0s_1 - 0s_2 - A_1$$

$$\text{STC } 2x_1 + x_2 - 6x_3 + A_1 = 20$$

$$6x_1 + 5x_2 + 10x_3 + s_1 = 76$$

$$8x_1 - 3x_2 + 6x_3 + s_2 = 50, \quad x_1, x_2, x_3, s_1, s_2, A_1 \geq 0$$

Phase 1 Write the Auxiliary equation

$$\text{Max } Z^* = -A_1$$

$$\text{STC } 2x_1 + x_2 - 6x_3 + A_1 = 20 \quad -(1)$$

$$6x_1 + 5x_2 + 10x_3 + s_1 = 76 \quad -(2)$$

$$8x_1 - 3x_2 + 6x_3 + s_2 = 50, \quad x_1, x_2, x_3, s_1, s_2, A_1 \geq 0$$

Basis	x_1	x_2	x_3	s_1	s_2	A_1	RHS	Ratio
A_1	2	1	-6	0	0	1	20	10
s_1	6	5	10	1	0	0	76	12.66
s_2	8	-3	6	0	1	0	50	6.25
Obj. Func	0	0	0	0	0	1	0	

Columns of the basic variables does not form the identity matrix hence eliminate 1 using row operation: $R'_4 \rightarrow R_4 - R_1$

Now apply simplex algorithm to solve the obtained LPP

Basis	x_1	x_2	x_3	s_1	s_2	A_1	RHS	Ratio
A_1	2	1	-6	0	0	1	20	10
s_1	6	5	10	1	0	0	76	12.66
s_2	8	-3	6	0	1	0	50	6.25
Obj. Func	-2	-1	6	0	0	0	-20	

Basis	x_1	x_2	x_3	s_1	s_2	A_1	RHS	Ratio
A_1	0	7/4	-15/2	0	-1/4	1	15/2	4.28
s_1	0	29/4	11/2	1	-3/4	0	77/2	5.3
x_1	1	-3/8	3/4	0	1/8	0	25/4	-
Obj. Func	0	-7/4	15/4	0	1/4	0		

Basis	x_1	x_2	x_3	s_1	s_2	RHS	Ratio
x_2	0	1	-30/7	0	-1/7	30/7	
s_1	0	0	256/7	1	2/7	52/7	
x_1	1	0	-6/7	0	1/14	55/7	
Obj. Func	0	0	0	0	0		

Since there are no negative numbers in the last row, table is in optimum state. Hence continue to phase 2

Phase 2

Remove the column A_1 and replace the costs of the original objective function in the last row of the table obtained in phase 1

Basis	x_1	x_2	x_3	s_1	s_2	RHS	Ratio
x_2	0	1	$-30/7$	0	$-1/7$	$30/7$	
s_1	0	0	$256/7$	1	$2/7$	$52/7$	
x_1	1	0	$-6/7$	0	$1/14$	$55/7$	
Obj. Func	-5	4	-3	0	0	0	

In Phase 2
 1. Remove A_i 's columns
 2. Keep all the rows except last row as it is and take the coefficients of the original obj function for the last row

Since x_1, x_2 and s_1 columns does not constitute the standard form, convert the table in to standard form, i.e., intersection of x_1 (and x_2, s_2) row with x_1 (and x_2, s_2) column =1 and all the elements above and below are 0 by performing row operation

$$R'_4 \rightarrow R_4 + 5R_3 - 4R_1$$

Basis	x_1	x_2	x_3	s_1	s_2	RHS	Ratio
x_2	0	1	$-30/7$	0	$-1/7$	$30/7$	
s_1	0	0	$256/7$	1	$2/7$	$52/7$	
x_1	1	0	$-6/7$	0	$1/14$	$55/7$	
Obj. Func	0	0	$69/7$	0	$13/14$	$155/7$	

Since there are no positive numbers in the last row, table is in optimum state. Hence the solution is

$$x_1 = 55/7, \quad x_2 = 30/7, \quad x_3 = 0, \quad \text{Max } Z = 155/7$$

Points to remember to solve LPP:

Simplex Method:	• Problems should be of type Max – Standard form • The basis should have the variables which are added in standard form • If the decision variables(x, y, z or x_i) are not present the basis then the Linear Programming Problem (LPP) is unsolvable • While doing row operation, do not multiply any constant to the row which is changing other than key row • Columns corresponding to Basic variables should form an identity matrix.
Big M Method	• All the rules of simplex method is applicable • Introduce artificial variables along with the surplus variables to $\geq$ constraints • Introduce only artificial variables to constraint equations • If an artificial variable remains in the first column at the optimal level of the simplex table, then the Linear Programming Problem (LPP) is unsolvable • Whenever an artificial variable happens to leave in the first column, remove the artificial variable column
Two-phase Method	• All the rules of simplex method is applicable • Consider only artificial variables in objective function in Phase 1 • Keeping all other rows replace costs of the original function in LPP and solve

Online Calculators

Graphical Method: <https://cbom.atozmath.com/CBOM/Simplex.aspx?q=gm>

Simplex Method or Big M method: <https://linprog.com/en/main-simplex-method>

Two phase simplex method: <https://www.pmcalculators.com/simplex-method-calculator/>